1. ESP32 2.8 Inch dispaly Instruction

Product Introduction

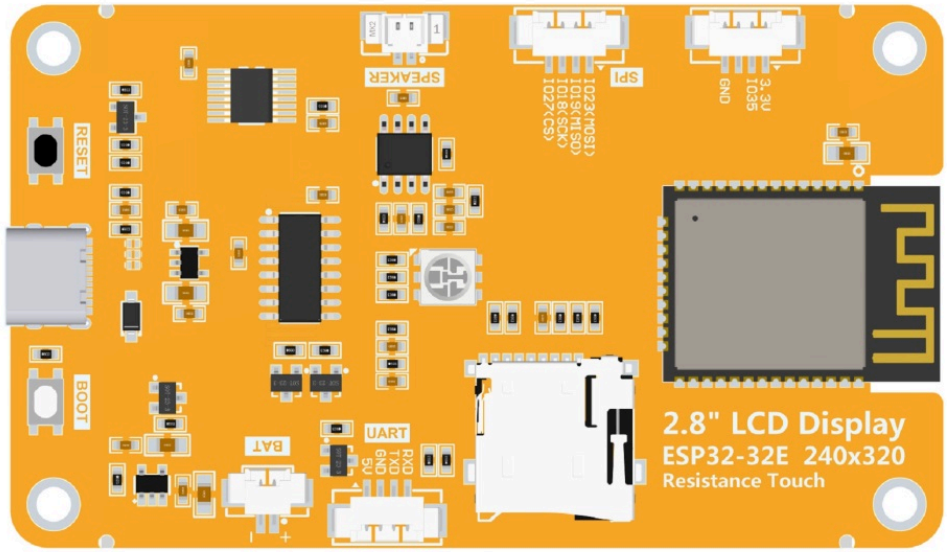
The E32R28T is a 2.8-inch display module controlled by the ESP32-WROOM-32E module, equipped with a resistive touch screen.

It adopts the ILI9341 display driver IC with a resolution of 240×320. It supports connections to various peripheral devices, including SPI devices, speakers, lithium batteries, MicroSD cards, serial ports and more.

Built-in touch capability greatly facilitates users to carry out various DIY projects, and it can meet users' development demands for the Internet of Things (IoT) and human-computer interaction functions.



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Accessory List

The module is supplied with two accessories: a resistive stylus pen for resistive touch screen operation, and a Type-C charging/data cable



PRODUCTPARAMETERS

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ESP32 main control parameters

Item	Parameter	Unit
Module	ESP32-WROOM-32E	-
CPU	ESP32-D0WD-V3, Xtensa Dual-Core 32-bit LX6 Microprocessor	-
Main Frequency	Max. 240MHz	-
Memory	448KB ROM + 520KB SRAM + 16KB RTC SRAM + 4MB External QSPI	-
Wi-Fi	2.4GHz, 802.11b/g/n standard	-
Bluetooth	Bluetooth V4.2 BR/EDR & Bluetooth LE	-
Operating Voltage	3.0 ~ 3.6	V
erating Temperatu	-40 ~ 85	℃

TFT Parameters

Item	Parameters	Unit
Panel Size	2.8	inch
Panel Type	TN TFT	-
Resolution	240xRGBx320	pixels
Active Area	43.20(W)x57.60(H)	mm
Number of Colors	Max: 262K(RGB666) Common: 65K(RGB565)	-
Driver IC	ILI9341V	-
Display Interface	4-Line SPI(Connect to ESP32)	-
Pixel Size	0.153(H)x0.153	mm
View Angle	12 0'CLOCK	deg
Brightness(TYP)	260	cd/m^2
Backlight Type	White LED*4	-
Operation Temperature	-10~50	℃
Storage Temperature	-20~60	℃

Touch Screen Parameters

Item	Parameters	Unit
Touch Active Area	2.8	inch
Touch Screen Type	Resistive touch screen	-
Touch Screen Resolution	240×320	pixels
Driver IC	XPT2046	-
Touch Screen Visual Area	45.20(W) × 59.45(H)	mm
Communication Interface	SPI	-
Material	ITO film + ITO glass	-
Operation Temperature	-10 ~ 60	℃
Storage Temperature	-20 ~ 70	℃

Size Parameters

Item	Parameters	Unit
Charging Voltage	Range: 4.2 to 6.5 Typical value: 5.0	V
Charging Current	Max. : 500 Module Actual value: 290	mA
Charging Saturation Voltage	4.24	W
Charging Temperature	Module maximum value: 62	℃
Rechargeable Battery Specifications	3.7V polymer lithium battery	-

Battery charging parameters

Item	Parameters	Unit
Charging Voltage	Range: 4.2 to 6.5 Typical value: 5.0	V
Charging Current	Max. : 500 Module Actual value: 290	mA
Charging Saturation Voltage	4.24	W
Charging Temperature	Module maximum value: 62	℃
Rechargeable Battery Specifications	3.7V polymer lithium battery	-

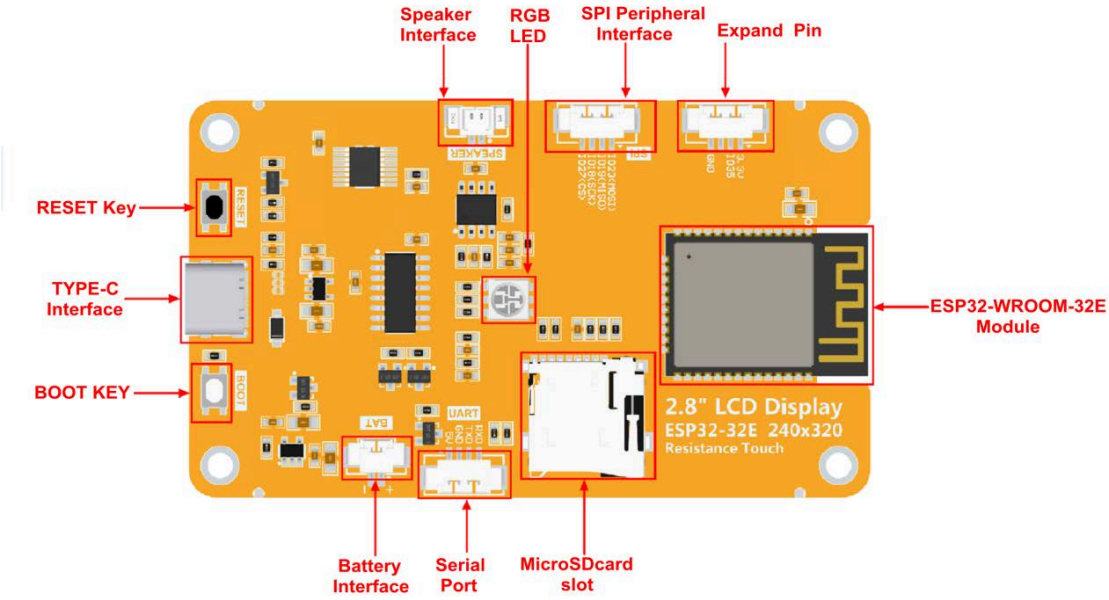
Electrical Parameters

Item	Parameters	Unit
Working Voltage	5.0	V
Backlight Current	75	mA
Total Current	ESP32 reset: 40 Only the display works: 130 Display, speaker, battery charging all work: 480	mA
Power	0.65 (Display only works)	W

PRODUCT INTERFACE

Interface Function Description

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Interface Name	Function Description
ESP32-WROOM-32E Module	Display module main control, control board peripheral and external peripheral.
MicroSD card slot	A MicroSD card is inserted to expand the storage space for storing large data such as characters, pictures, and audio files.
RGB LED	Contains red, green, blue three colors of LED lights, each light can be controlled by IO, used to indicate the status.
Serial port	1.25mm 4P connector.It can be used for serial debugging, downloading, and communication. An external USB to serial port module is required.
Battery interface	1.25mm 2P connector.Used to connect 3.7V polymer lithium battery, charge the battery through the battery charge management circuit, can also be used for battery power supply. Note the positive and negative terminals of the interface.
BOOT Key	Used to enter download mode or key test. Press and hold this button to power on and release it to enter the download mode. Alternatively, press and hold this button and then press the RESET button to release the RESET button and then release this button to enter the download mode. When you do not need to enter the download mode, this button can be used as a common button.
Type-C Interface	Used for module power supply and download programs. This interface is connected to the one-click download circuit on the module, which can automatically enter the download mode (without pressing the BOOT key).
RESET Key	Used for ESP32 master control and LCD reset, level reset after pressing.
Expand pin	1.25mm 2P connector. Lead out GND, 3.3V, and IO35 pins, of which IO35 can only be used as the input pin.
Speaker interface	1.25mm 2P connector. Used to access the speaker to play audio, this interface is connected to the audio power amplifier circuit.
SPI peripheral interface	1.25mm 4P connector. Used to connect devices that use SPI communication. This SPI interface is shared with MicroSD. It can be used as ordinary IO.

ESP32 pin allocation

Onboard Device	ESP32 Connect pins	Pin description of onboard equipment
LCD	IO15	LCD screen selection control signal, low level effective
	IO2	LCD command/data selection control signalHigh Level:data,low Level:command
	IO14	LCD SPI bus clock signal
	IO13	LCD SPI bus write data signal
	IO12	LCD SPI bus read data signal
	EN	LCD reset control signal, low level reset(share reset pin with ESP32-32E master)
	IO21	LCD backlight control signal(high level backlight on, low level backlight off)
Resistive touch screen	IO25	Resistive touch screen SPI bus clock signal
	IO32	Resistive touch screen SPI bus write data signal
	IO39	Resistive touch screen SPI bus read data signal
	IO33	Resistive touch screen chip selection control signal,low level effective
	IO36	Resistive touch screen touch interrupt signal,generates touch when input low level to master
RGB three-color light	IO22	Red LED light(common anode, low level on, high level off)
	IO16	Green LED light(common anode, low level on, high level off)
	IO17	Blue LED light(common anode, low level on, high level off)
MicroSD Card	IO5	SD card select signal,low level effective
	IO23	SD card SPI bus write data signal(shared by MicroSD card and SPI peripheral)
	IO18	SD card SPI bus clock signal(shared by MicroSD card and SPI peripheral)
	IO19	SD card SPI bus read data signal(shared by MicroSD card and SPI peripheral)
Audio	IO4	Audio enable signal,low level enable,high level disable
	IO26	Audio signal DAC output signal
KEY	IO0	Download mode Select button(Press and hold the button to power on, then release to enter download mode)
	EN	ESP32-32E reset button, low level reset(shared with LCD reset)
Serial Port	RXD0(IO3)	ESP32-32E serial port receiving signal(if the serial port is not used, it can be used as ordinary IO)
	TXD0(IO1)	ESP32-32E serial port sends signals(if the serial port is not used, it can be used as ordinary IO)
Battery	IO34	Battery voltage ADC value Get Signal (input)
SPI peripheral	IO27	SPI peripheral chip selection signal, low level effective(if the SPI device is not used, it can be used for ordinary IO)
	IO18	SPI bus clock pin for SPI peripherals (SPI peripherals are shared with MicroSD cards, if SPI devices or SD cards are not used, ordinary IO can be used)
	IO19	The SPI bus read data pin of the SPI peripheral (SPI peripherals are shared with MicroSD cards, if SPI devices or SD cards are not used, ordinary IO can be used)
	IO23	The SPI bus of the SPI peripheral writes data pins (SPI peripherals are shared with MicroSD cards, if SPI devices or SD cards are not used, ordinary IO can be used)
NC	IO35	It can only be used as input IO

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